Robin Connor Blalock

Murfreesboro, TN | connor.blalock77@gmail.com
cblalock.github.io/portfolio | github.com/cblalock | linkedin.com/in/robin-blalock

Projects

Autonomous Options Trading System | Python, FastAPI, Claude API, Alpaca Markets, PostgreSQL, SQLite, JavaScript | setupscan.app

- Architected a three-part trading system end-to-end: a live scanning SaaS product, an autonomous Claude-driven trading agent, and a performance dashboard that analyzes the agent's own decisions.
- Shipped Setup Scanner (setupscan.app) as a paid subscription product: a FastAPI and PostgreSQL platform scanning NYSE and NASDAQ equities across 15 technical indicators and 5 timeframes, with Stripe billing and JWT auth.
- Built a Claude-powered agent that consumes the scanner's live signals through its own API, autonomously executes options trades via Alpaca, and journaled 190+ closed paper trades, full entry and exit reasoning included, to a SQLite audit trail.
- Engineered a cost-constrained agent runtime, using deterministic signal pre-filtering and prompt caching to keep Claude spend low, wrapped in a model-independent safety layer (hard stop-losses, broker-liquidation handling, re-entry cooldowns).
- Analyzed the agent's trade history with a SQL and Pandas pipeline and two dashboards (Tableau and a hand-coded web version), finding a setting that kept 13 of 15 indicators from firing, then testing and reverting the change.
- Removed every guardrail as a final experiment; average loss per trade rose from $8.62 to $24.37, showing the guardrails were limiting losses rather than hiding an edge.

Private AI Assistant | Ansible, Docker, Ollama, OpenWebUI, Ubuntu | github.com/cblalock/private-ai-assistant

- Authored an Ansible playbook that provisions a fully operational private AI assistant from a single command.
- Designed around a real-world use case: organizations that need an internal LLM without routing sensitive data through third-party servers.
- Automated Docker container orchestration, OpenWebUI deployment, and systemd service configuration across Ubuntu hosts.

UAB Game Launcher | React, Node.js, Express, SQLite, JavaScript | github.com/cblalock/CS499GameLauncher

- Delivered a full-stack single-page web application as a Senior Capstone project for UAB's Biology Department, a real external stakeholder, and won 1st place among capstone teams.
- Designed and built a REST API with Node.js and Express, backed by SQLite, to manage user accounts, leaderboards, and high scores across a library of launchable educational games.
- Built the React front end end-to-end, from game-library browsing to account and score views, consuming the API directly.

Education

- **University of Alabama at Birmingham** | BA, Computer Science; Minor in Psychology | 2025
- **Calhoun Community College** | AS, Science | 2020
- **US Army Technical Training** | 25Q Multichannel Transmission Systems Operator | VSAT Terminals | 2014-2018

Technical Skills

- **Languages:** Python, JavaScript, Java, C, C++, SQL, HTML/CSS
- **Frameworks & Libraries:** React, Node.js, Express, FastAPI, Pandas, NumPy
- **Tools & Platforms:** Claude API, LLM Agents, Git, Docker, Ansible, Linux, Unreal Engine 5
- **Concepts:** Data Structures, Algorithms, REST APIs, Real-Time Systems, Client-Server Architecture

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Professional Experience

Yard Robotics | Technician | 2026 – Present

- Diagnosed robotic hardware and connectivity issues using a structured LED status-code system (e.g., charging state, GPS signal status), performing remediation such as battery and part replacement and CPU resets.
- Contributed to yard mapping for autonomous navigation by manually driving robots through unmapped areas to generate path-planning data.

United States Army | 25Q | 2014 – 2018

- Operated and maintained secure multi-channel communication and satellite (VSAT) systems in mission-critical environments.
- Commissioned newly fielded VSAT terminals from a completely unconfigured, factory-default state, using extensive command-line configuration to bring new unit equipment into operational service.
- Configured network bridges, tunnels, and routing via CLI, and performed satellite antenna alignment, to establish and maintain secure uplink connectivity between VSAT terminals and the command satellite.
- Supported communications security (COMSEC) operations and performed complex network troubleshooting.
- Conducted hardware diagnostics on communication equipment, vehicles, and power systems.
- Held Secret Clearance.

Mazda Toyota Manufacturing | Team Member | 2023 – 2024

- Maintained a zero-defect rate across multiple production processes; cross-trained on multiple lines and assisted with scheduling and process optimization.

Weather Wise Exteriors | Installer | 2018 – 2020

University of Alabama at Birmingham | Full Time Student | 2020-2022, 2025